

Skills Summary

Unit Fractions in Area Models

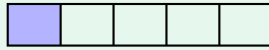
Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the Unit Fraction in a Model

Rule: a **unit fraction** is one equal piece of a whole. Its top number is always 1; its bottom number is how many equal pieces make the whole.

one piece out of d equal pieces = $\frac{1}{d}$. The shape does not matter — bar, circle or grid — only the count of equal pieces does.

Example: One piece of this bar is shaded. Name the unit fraction.



Step 1. The question asks for one piece of a whole, so I need a unit fraction — the top number is fixed at one and I only have to count the equal pieces.

Step 2. Counting the pieces across the bar gives 5 equal pieces, so the bottom number is 5.
one shaded piece = $\frac{1}{5}$

Try it: A pizza is cut into 4 equal slices. What unit fraction is one slice?

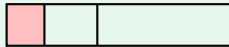
check: $\frac{4}{1}$

Skill 2 — The Pieces Must Be Equal

Watch the word “equal.” Counting pieces is not enough. A shape cut into three pieces only shows *thirds* when all three pieces are the same size.

Test it: could the pieces be swapped and still cover the same amount? If not, they are not equal parts and no piece is a unit fraction of that whole.

Example: Does the shaded piece show one third?



Step 1. Before naming any fraction I check the equal-parts condition, because a unit fraction is only defined when every piece is the same size.

Step 2. The three pieces here are clearly different widths — the shaded one is the narrowest — so this model does not show thirds at all, and the shaded piece has no unit-fraction name.

Answer: no. Three pieces, but not three *equal* pieces.

Try it: A strip is cut into two pieces, one much longer than the other. Does the short piece show one half? Say why.

check: no — halves must be two pieces of the same size

Skill 3 — Comparing Unit Fractions

Rule: when two wholes are the *same size*, more pieces means each piece is *smaller*.

So for unit fractions, the bigger bottom number names the smaller piece: $\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{6} > \frac{1}{8}$.

Example: Two same-size bars: one cut into 4 pieces, one cut into 8. Which shaded piece is bigger?

Step 1. Both wholes are the same size, so I can compare the pieces just by comparing how many pieces each whole was cut into.

Step 2. Cutting the same bar into 8 pieces makes each piece smaller than cutting it into 4, so the fourth is the bigger piece.

$$\frac{1}{4} > \frac{1}{8}$$

Try it: Same-size bars, one cut into 2 pieces and one into 6. Write $>$, $<$ or $=$ between $\frac{1}{2}$ and $\frac{1}{6}$.

check: $\frac{9}{1} < \frac{2}{1}$